MALL CUSTOMER SEGMENTATION & ALGORITHM COMPARISON DASHBOARD

A PROJECT REPORT

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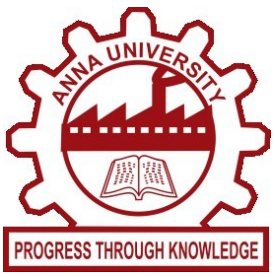
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in

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BONAFIDE CERTIFICATE

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ABSTRACT

This Principles of Data Science mini project presents an interactive customer segmentation and algorithm comparison system using the standard Kaggle Mall_Customers.csv dataset containing 200 customer records with Age, Gender, Annual Income in k$, and Spending Score 1-100. The workflow begins with exploratory data analysis and feature preparation, followed by StandardScaler-based normalization so that Euclidean-distance-driven clustering is not dominated by Annual Income. Three unsupervised clustering approaches are studied: two-dimensional K-Means using Annual Income and Spending Score with k=5, three-dimensional K-Means using Age, Annual Income and Spending Score with k=6, and Agglomerative Hierarchical Clustering using Ward's linkage on Annual Income and Spending Score. The Elbow Method is used to identify the 2D K-Means choice of five clusters, while the hierarchical model is interpreted using a SciPy dendrogram. The resulting segments are mapped to business-oriented personas such as Target Audience, Careless Spenders, Cautious Buyers, Sensible Savers, and Standard Customers. Model quality is compared using the Silhouette Score and Davies-Bouldin Index. The complete workflow is deployed as a multi-tab Streamlit dashboard containing dynamic k controls, real-time metric updates, Matplotlib visualizations, and an interactive Plotly 3D scatter plot. The project demonstrates how unsupervised learning and visualization can convert customer data into interpretable segments for exploratory marketing analysis.

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CHAPTER 1

INTRODUCTION

Customer segmentation is an important application of unsupervised learning in which customers are grouped according to similar behavioral or demographic characteristics without requiring predefined class labels. In retail and marketing, segmentation can support targeted campaigns, customer profiling, product recommendations, and resource allocation. The Mall_Customers.csv dataset is a compact and widely used dataset for demonstrating this concept because it combines demographic information with spending behavior.

1.1 Background

The dataset contains 200 customer records and the attributes Age, Gender, Annual Income in k$, and Spending Score 1-100. For the principal 2D segmentation study, Annual Income and Spending Score are selected because they provide an interpretable business view of purchasing power and spending behavior. Age is introduced in a separate 3D K-Means experiment to examine whether an apparently broad average-spending group can be divided further when demographic information is considered.

1.2 Motivation

A single clustering algorithm can produce useful groups, but its result depends on feature selection, distance scale, linkage strategy, and the selected number of clusters. This project is motivated by the need to compare clustering approaches rather than relying on one model. A dashboard also makes the analysis easier to inspect by allowing the user to change k, observe metrics, and explore the resulting clusters interactively.

1.3 Objectives

The objectives are to inspect and preprocess the Mall Customers dataset; apply StandardScaler to the selected numerical features; determine a suitable K-Means configuration using the Elbow Method; build 2D and 3D K-Means models; construct an Agglomerative Hierarchical Clustering model using Ward's linkage; visualize the hierarchical structure using a SciPy dendrogram; compare algorithms using Silhouette Score and Davies-Bouldin Index; interpret the clusters as meaningful customer personas; and deploy the complete workflow through a Streamlit dashboard.

1.4 Problem Statement

Retail customer data often contains mixed demographic and behavioral information, making it difficult to identify actionable groups through manual inspection alone. Customers with similar income may exhibit very different spending behavior, while customers with similar spending scores may differ by age or income. The project therefore studies how unsupervised clustering can reveal customer groups and how different clustering algorithms can be compared through objective internal validation metrics.

1.5 Scope of the Project

The scope covers exploratory analysis of the Mall_Customers.csv dataset, numerical feature scaling, K-Means clustering in two and three dimensions, hierarchical clustering, dendrogram analysis, internal cluster validation, business-oriented cluster interpretation, and Streamlit deployment. The dashboard is an academic analytical prototype; the resulting personas should be treated as exploratory segments rather than permanent customer labels..

CHAPTER 2

LITERATURE SURVEY (EXISTING SYSTEM)

Customer segmentation has traditionally been performed using demographic rules, manually designed categories, or statistical grouping methods. Unsupervised machine learning extends this process by allowing patterns in unlabeled observations to determine the groups. K-Means is one of the most commonly introduced partitioning algorithms because it is computationally simple, interpretable, and effective for compact numeric datasets.

K-Means assigns observations to the nearest centroid and iteratively updates the centroids to reduce within-cluster squared distance. The Elbow Method uses WCSS, or within-cluster sum of squares, to study how this objective changes as k increases. Because K-Means is distance-based, feature scaling is important when variables are measured on different numerical ranges. StandardScaler transforms each selected feature using its mean and standard deviation so that no single feature dominates the distance calculation merely because of its units.

Hierarchical clustering provides a complementary perspective because it builds a hierarchy of nested groups instead of directly requiring a final partition. Agglomerative clustering begins with individual observations and repeatedly merges clusters. Ward's linkage chooses merges that produce the smallest increase in within-cluster variance. The resulting hierarchy can be visualized using a dendrogram, allowing the analyst to inspect possible cut levels and cluster structure.

Internal validation metrics are useful when ground-truth customer labels do not exist. The Silhouette Score considers how close an observation is to its

own cluster compared with neighboring clusters, with larger values generally indicating better separation and cohesion. The Davies-Bouldin Index measures the similarity between clusters using within-cluster scatter and between-cluster separation, with smaller values generally preferred. Using both measures gives a more balanced view than relying on a single score. The proposed project combines these concepts with interactive visualization so that numerical evaluation and business interpretation can be considered together.

CHAPTER 3

ARCHITECTURE OF THE MALL CUSTOMER SEGMENTATION SYSTEM

The proposed architecture follows a complete data-science pipeline from raw customer records to interactive decision support. The input dataset is inspected and cleaned before numerical features are scaled. The clustering layer contains 2D K-Means, 3D K-Means, and Ward-based Agglomerative Clustering. The evaluation layer computes Silhouette Score and Davies-Bouldin Index, while the interpretation layer maps the 2D segments to business personas. Finally, Streamlit exposes the analysis through interactive tabs, controls, metrics, and plots.

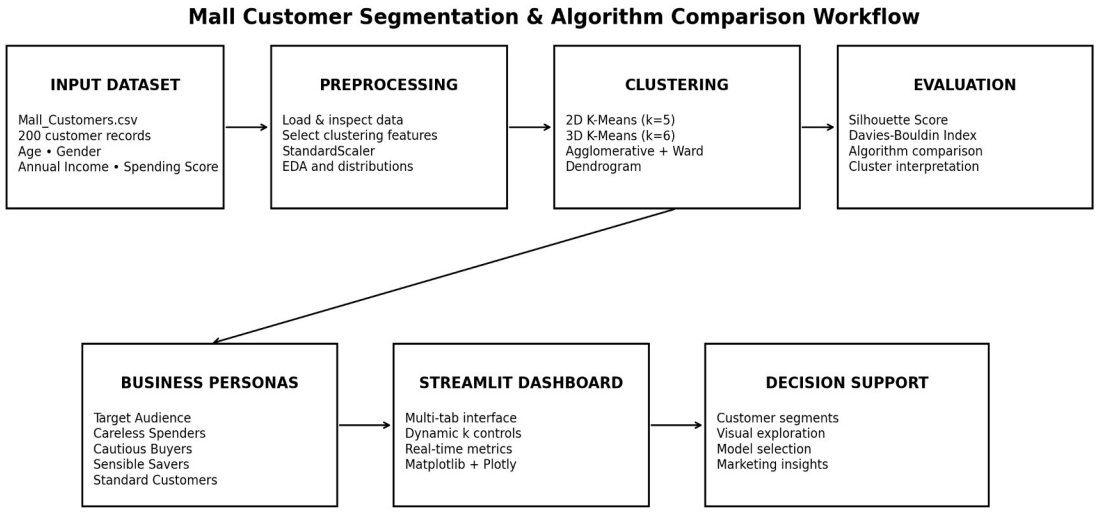


Figure 1: Architecture of the Mall Customer Segmentation Workflow

Input Data Layer

The input is the standard Kaggle Mall_Customers.csv dataset containing 200 records. The main attributes are CustomerID, Gender, Age, Annual Income (k$), and Spending Score (1-100). CustomerID is an identifier rather than a clustering feature. Gender can be retained for exploratory analysis, while the principal clustering experiments use the numerical variables needed for the selected feature spaces.

Preprocessing Layer

The preprocessing stage checks the dataset structure, missing values, data types, duplicate observations, and feature ranges. For clustering, the selected numerical features are transformed using StandardScaler. This is important because Annual Income, Age, and Spending Score have different numerical scales, and distance-based algorithms can otherwise be disproportionately influenced by the variable with the largest magnitude.

Clustering Layer

The clustering layer contains three experiments. The first is 2D K-Means on Annual Income and Spending Score, where the Elbow Method identifies k=5 for the principal segmentation. The second is 3D K-Means on Age, Annual Income, and Spending Score using k=6 to demonstrate how age can split an otherwise broad average-spending group. The third is Agglomerative Hierarchical Clustering on Annual Income and Spending Score using Ward's linkage, with the SciPy dendrogram used to inspect the hierarchy and validate a five-cluster interpretation.

Evaluation and Visualization Layer

The evaluation layer computes the Silhouette Score and Davies-Bouldin Index for the compared clustering configurations. Matplotlib is used for standard statistical and clustering plots, while Plotly provides an interactive 3D scatter visualization. The dashboard can expose these metrics in real time as the user changes the selected k value.

Application Layer

The Streamlit application organizes the workflow into multiple tabs so that dataset inspection, exploratory analysis, clustering, model comparison, and 3D visualization can be viewed without changing the underlying code. Dynamic controls allow the user to experiment with k and immediately inspect the effect on the cluster assignments and internal validation metrics.

CHAPTER 4

SYSTEM ANALYSIS

4.1 Existing System

A basic customer-analysis workflow often relies on manual grouping or a single clustering algorithm executed in a notebook. Although such approaches can demonstrate the algorithm, they make it difficult for a non-technical user to compare different cluster structures, change hyperparameters, and inspect validation metrics in one place. Static plots also provide limited interaction with three-dimensional customer relationships.

4.2 Proposed System

The proposed system integrates exploratory data analysis, feature scaling, three clustering strategies, internal evaluation, business persona mapping, and Streamlit deployment into one reproducible pipeline. The system is designed to answer both technical and business questions: how the clusters are formed, how the algorithms differ, how well the clusters are separated, and what customer behavior each group represents.

4.3 System Requirements

Hardware requirements include a standard computer with sufficient memory to load the dataset and execute Python-based clustering and visualization. A GPU is not required for this project. Software requirements include Python, Pandas, NumPy, Scikit-Learn, SciPy, Matplotlib, Seaborn, Plotly, and Streamlit. Functional requirements include CSV loading, EDA, scaling, clustering, metric calculation, visualization, interactive controls, and dashboard execution. Non-functional requirements include reproducibility, clear visual interpretation, responsive interaction, and separation of data processing from presentation.

CHAPTER 5

MODULES DESCRIPTION

5.1 Dataset Inspection Module

The Dataset Inspection Module loads Mall_Customers.csv using Pandas and verifies the number of records, column names, data types, missing values, and basic descriptive statistics. Exploratory visualizations can be used to inspect distributions and relationships between Age, Annual Income, and Spending Score before clustering. The module also separates identifier fields from features that are meaningful for distance-based analysis.

5.2 Data Preprocessing and Scaling

The Data Preprocessing Module selects the numerical feature matrix required by each experiment and applies StandardScaler. The fitted scaler transforms the selected features into a standardized space. Scaling is performed before distance-based clustering so that Annual Income does not dominate the Euclidean distance because of its larger numerical magnitude.

5.3 2D K-Means Clustering

The 2D K-Means module uses Annual Income and Spending Score. Candidate values of k are evaluated through WCSS and the Elbow Method, with k=5 selected for the principal segmentation. The resulting clusters are interpreted using five business personas: Target Audience for high-income/high-spending customers, Careless Spenders for relatively lower-income but high-spending customers, Cautious Buyers for higher-income but lower-spending customers, Sensible Savers for lower-income/low-spending customers, and Standard Customers for the more average region.

5.4 3D K-Means Clustering

The 3D K-Means module extends the feature space to Age, Annual Income, and Spending Score. A configuration of k=6 is used to demonstrate how adding Age can create an additional separation inside a broad average-spending population. The three-dimensional structure is explored through a Plotly scatter cube so that users can rotate the view and inspect the relationship between demographic and behavioral variables.

5.5 Agglomerative Hierarchical Clustering

The hierarchical module uses Annual Income and Spending Score with Ward's linkage. Unlike K-Means, the method begins with individual observations and repeatedly merges the closest groups according to the Ward criterion. The SciPy dendrogram provides a visual representation of the merging process and supports a five-cluster cut without directly beginning from a predefined centroid configuration.

5.6 Model Evaluation

The evaluation module benchmarks the clustering configurations using Silhouette Score and Davies-Bouldin Index. The Silhouette Score rewards compact clusters that are well separated from neighboring clusters, whereas the Davies-Bouldin Index favors low within-cluster scatter and strong separation. Both metrics are displayed as numerical outputs in the dashboard so that algorithm choices can be evaluated using the same validation criteria.

5.7 Streamlit Dashboard

The dashboard module wraps the full pipeline into a multi-tab Streamlit application. It provides dataset views, EDA plots, cluster visualizations, model metrics, dynamic k sliders, and an interactive Plotly 3D view. The interface is designed so that changing a hyperparameter triggers the relevant calculation and updates the displayed visualization and validation metrics.

CHAPTER 6

IMPLEMENTATION

The implementation begins by loading Mall_Customers.csv into a Pandas DataFrame and inspecting its shape, columns, missing values, and descriptive statistics. Exploratory analysis is then performed using Seaborn and Matplotlib to understand the distributions of Age, Annual Income, and Spending Score. The clustering feature matrices are created according to the selected experiment and transformed with StandardScaler.

For 2D K-Means, the implementation evaluates multiple values of k and records the WCSS used by the Elbow Method. The chosen configuration uses k=5. After fitting the model, each customer receives a cluster label, and the cluster centroids are examined to assign interpretable business personas. The 3D experiment uses Age together with Annual Income and Spending Score and operates with k=6. The resulting labels are visualized using an interactive Plotly scatter plot.

The hierarchical implementation uses SciPy's linkage and dendrogram functionality with Ward's method. The dendrogram is used to inspect the merging structure and identify a reasonable five-cluster cut. For each clustering result, Silhouette Score and Davies-Bouldin Index are calculated from the standardized feature representation used by the respective model.

The Streamlit application connects the analytical modules to a user interface containing multiple tabs and controls. The code is organized so that data preparation, clustering, evaluation, and plotting remain logically separate. Matplotlib and Seaborn handle standard two-dimensional visualizations, while Plotly handles the interactive three-dimensional view. This modular

implementation makes it easier to test individual components and extend the dashboard with additional clustering algorithms.

CHAPTER 7

RESULTS AND DISCUSSION

The project produces three complementary views of the same customer population. The 2D K-Means model provides the main business-oriented segmentation and identifies five groups using the Annual Income and Spending Score feature space. The clusters can be interpreted as Target Audience, Careless Spenders, Cautious Buyers, Sensible Savers, and Standard Customers. This mapping converts numerical cluster labels into descriptions that are easier to communicate in a marketing context.

The 3D K-Means experiment demonstrates the effect of introducing Age into the feature space. Using k=6 allows the model to split a broad average-spending region into more detailed demographic groups. This highlights an important principle of segmentation: the choice of features changes the meaning and granularity of the resulting clusters. A segment that looks homogeneous in two dimensions may contain meaningful internal differences when an additional variable is considered.

The hierarchical experiment provides a different view of the same income-versus-spending relationship. The Ward dendrogram shows how observations merge as the distance threshold increases and visually supports a five-cluster interpretation. Unlike K-Means, the dendrogram allows the analyst to inspect the full hierarchy before deciding where to make the cut.

The final comparison is performed using Silhouette Score and Davies-Bouldin Index rather than subjective visual quality alone. The dashboard presents these metrics alongside the plots so that a technically stronger clustering configuration can be considered together with its business interpretability.

Exact metric values should be taken directly from the completed implementation when the final report is submitted; this report does not invent unverified numerical results.

CHAPTER 8

FUTURE ENHANCEMENTS

The project can be extended by testing additional clustering algorithms such as DBSCAN, Gaussian Mixture Models, and Spectral Clustering. More systematic hyperparameter search can also be added so that candidate k values and algorithm settings are evaluated automatically rather than manually through dashboard controls. Alternative internal metrics such as Calinski-Harabasz Score can provide another perspective on cluster compactness and separation.

Future versions can incorporate categorical information such as Gender using suitable encoding strategies, add temporal or transactional customer behavior when available, and support model persistence so that the chosen segmentation can be reused without retraining. The Streamlit dashboard can also be extended with downloadable cluster reports, persona summaries, recommendation rules, and interactive filtering. For a production-oriented system, cluster stability, drift monitoring, and validation on larger and more representative customer datasets would be important.

CHAPTER 9

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